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Are There Environmental Benefits from Driving Electric Vehicles? The Importance of Local Factors†

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We combine a theoretical discrete-choice model of vehicle purchases, an econometric analysis of electricity emissions, and the AP2 air pollution model to estimate the geographic variation in the environmental benefits from driving electric vehicles. The second-best electric vehicle purchase subsidy ranges from \$2,785 in California to −\$4,964 in North Dakota, with a mean of −\$1,095. Ninety percent of local environmental externalities from driving electric vehicles in one state are exported to others, implying they may be subsidized locally, even when the environmental benefits are negative overall. Geographically differentiated subsidies can reduce deadweight loss, but only modestly. (*JEL* D12, D62, H23, L62, Q53, Q54, R11)

For a variety of reasons, including technological advances, environmental concerns, and entrepreneurial audacity, the market for pure electric vehicles, which was moribund for more than a century, is poised for a dramatic revival.[1] Several models are already selling in considerable volumes, the portfolio of electric vehicles is beginning to span the vehicle choice set, and almost all major manufacturers are bringing new models to the market. The Federal Government is encouraging these developments by providing a significant subsidy for the purchase of an electric vehicle, and some states augment the federal policy with their own additional subsidy.[2]

Proponents of these subsidies argue electric vehicles generate a range of short-term and long-term benefits such as reduced environmental impacts, innovation

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[1] http://energy.gov/articles/history-electric-car (accessed October 17, 2016).

[2] Internal Revenue Code Section 30D (Notice 2009-89) provides a tax credit of up to \$7,500.

spillovers, and reduced reliance on imported oil.[3] In this paper we analyze whether electric vehicles do indeed generate short-term environmental benefits by examining air pollution damages from driving gasoline vehicles and charging electric vehicles. In particular, we focus on the importance of local factors by including global and local pollution, spatial heterogeneity of damages, pollution export across political jurisdictions, and policy that may vary by location.

Three main considerations motivate our analysis. First, prior studies of electric vehicles have focused on calculating the emissions of electric vehicles but have not had a conceptual framework for analyzing electric vehicle subsidies.[4] We analyze a model of vehicle choice, which gives us the theoretically sound and intuitive result that the subsidy should be equal to the difference in lifetime damages between an electric vehicle and a gasoline vehicle. Our theoretical framework also allows us to address additional policy questions regarding the best policies for different jurisdictional levels and the welfare gains from policy differentiation.[5]

Second, despite being treated by regulators as "zero emission vehicles," electric vehicles are not necessarily emissions free (see, for example, National Academy of Sciences 2010). In 2014, the US Department of Energy reported that nearly 70 percent of electricity generated in the United States is produced by burning coal and natural gas. In many locations, the comparison between a gasoline vehicle and an electric one is really a comparison between burning gasoline or a mix of coal and natural gas to move the vehicle. However, average emissions of regional power plants can be a misleading indicator of the environmental impact of electric cars because all power plants do not respond proportionally to an increase in electricity usage and because electricity flows do not respect regional (e.g., state) boundaries.[6] To assess the emissions from charging an electric vehicle, we use an econometric model to estimate the effect of charging an electric vehicle on the marginal emissions of multiple pollutants at each power plant.[7]

Third, there are significant physical differences between emissions from gasoline and electric vehicles. This is due to the distributed nature of the electricity grid, the height at which emissions occur, and the chemistry of fuel combustion. As a result, pollutants and emissions rates may be spatially distinct even if gasoline and electric vehicles are driven in the same place. For local pollutants, an additional problem is that the same vehicle driven in different places leads to different damages. For this reason, many prior studies consider only carbon dioxide.[8] We use an integrated assessment model to value damages across local and global pollutants for both electric and gasoline vehicles.[9]

[3] http://energy.gov/eere/vehicles/ev-everywhere-grand-challenge-does-10-year-vision-plug-electric-vehicles (accessed October 17, 2016).

[4] See, for example, Graff Zivin, Kotchen, and Mansur (2014) and Michalek et al. (2011).

[5] Examples of theoretical discrete choice transportation models include De Borger (2001); De Borger and Mayeres (2007); and Parry and Small (2005). Differentiated policy is analyzed by Weitzman (1974); Mendelsohn (1986); Banzhaf and Chupp (2012); and Fowlie and Muller (2013).

[6] The EPA's calculated CO_2 emissions rates for electric vehicles (www.fueleconomy.gov) are regional averages.

[7] This builds on Graff Zivin, Kotchen, and Mansur (2014) and Holland and Mansur (2008).

[8] See, for example, Graff Zivin, Kotchen, and Mansur (2014) and Archsmith, Kendall, and Rapson (2015).

[9] Previous air pollution integrated assessment research includes Mendelsohn (1980); Burtraw et al. (1998); Mauzerall et al. (2005); Tong et al. (2006); Fann, Fulcher, and Hubbell (2009); Levy, Baxter, and Schwartz (2009); Muller and Mendelsohn (2009); and Henry, Muller, and Mendelsohn (2011). We model both ground-level emissions

Addressing these three considerations yields a powerful modeling framework for analyzing electric vehicle policy. In particular, our study is the first to consider the geographic variation in damages from both local and global pollutants emitted by both gasoline and electric vehicles and to tie this variation to a choice model.[10] This framework enables us not only to evaluate the environmental benefit of electric cars, but also to address questions of political economy and fiscal federalism.

Our first set of results documents the considerable heterogeneity in the environmental benefits of an electric vehicle relative to a gasoline vehicle. These benefits can be large and positive, large and negative, or negligible, depending on the location. For example, California has relatively large damages from gasoline vehicles and a relatively clean electric grid, which implies large positive environmental benefits of an electric vehicle. These conditions are reversed in North Dakota. The variation in the sign of the environmental benefits stems almost entirely from local air pollution. If we account only for greenhouse gases, then electric vehicles are superior to gasoline vehicles almost everywhere. Using our model, we determine the welfare maximizing subsidies on electric vehicle purchases. We refer to these subsidies as second-best, but we stress that they only account for the relative differences in environmental impacts from driving electric and gasoline vehicles in the short run. Even in locations like California, subsidy values are significantly less than the current federal subsidy. The national average subsidy for the purchase of an electric vehicle is estimated to be −\$1,095. Thus, on average in the United States, the second-best purchase policy is a tax, not a subsidy.

Our second set of results shows the remarkable degree to which electric vehicles driven in one location lead to environmental externalities in other locations. At the state level, 91 percent of local pollution damages from driving an electric vehicle are exported to states other than the state in which the vehicle is driven. In contrast, only 19 percent of local pollution damages from driving a gasoline vehicle are exported to other states. This discrepancy casts doubt on the efficacy of policy selected by local regulators. It is not obvious whether a given state will consider full damages (damages across all states), or only native damages (those damages which actually occur in the given state) when setting policy. Moreover, state regulators face incentives in current air pollution policy that emphasize within-state consequences of emissions. The National Ambient Air Quality Standards (NAAQS) emphasize compliance with ambient pollution limits within states. Although there are constraints on the extent of exported pollution, especially from power plants, the NAAQS clearly encourage local compliance. This leads state regulators to focus on in-state damage and hence prefer a technology that exports pollution to other regions. The difference between using full and native damages in determining the second-best subsidy may be considerable. Accounting for full damages, the second-best subsidy is positive in 11 states. Accounting for only native damages, the second-best subsidy is positive in 32 states.

and power plant emissions throughout the contiguous United States. In contrast to prior work, we report damages within the county of emission, within the state of emission, and in total (across all receptors).

[10] Babaee, Nagpure, and DeCarolis (2014); Graff Zivin, Kotchen, and Mansur (2014); Michalek et al. (2011); and Tessum, Hill, and Marshall (2014) analyze the benefits of electric vehicles at the aggregate level. Li et al. (forthcoming) consider variation in damages from electric vehicles but assume uniform damages from gasoline vehicles. Grissom (2013) considers variation in damages from gasoline vehicles but does not account for local pollution from electric vehicle charging. Archsmith, Kendall, and Rapson (2015) assess the life-cycle greenhouse gas (GHG) benefits from electric vehicles.

The final set of results assesses the deadweight loss of various policies as well as the welfare gains from differentiated policy. Our theoretical analysis reveals that the welfare gains from differentiated subsidies depend on the higher order moments of the distribution of environmental benefits. Calibrating this model gives us an estimate of the magnitude of these gains. For electric vehicle subsidies, we find large deadweight loss and small welfare gains from differentiation. For taxes on gasoline miles, we find small (or zero) deadweight loss and larger welfare gains from differentiation.

There are several important caveats to our calculation of the environmental benefits of electric vehicles. First, it only captures air pollution emissions associated with driving or charging the vehicles. It does not account for "upstream" environmental externalities associated with producing either fuels or vehicles. Second, it is based on the electricity grid in the years 2010–2012 and current gasoline vehicle technology.[11] Over time, both the grid and gasoline vehicles may become cleaner. Third, it depends on marginal emissions from an increase in the demand for electric power to charge electric vehicles. This may not be appropriate when electric vehicles comprise a substantial fraction of the vehicle fleet. Fourth, it ignores preexisting environmental polices such as the Corporate Average Fuel Economy (CAFE) standards and cap-and-trade markets for various local pollutants. For each of these caveats, we consider the degree to which they affect our calculated environmental benefits. For example, accounting for CAFE standards leads to an additional environmental cost of \$1,555 per vehicle, whereas if there were a binding cap on SO_2 the additional environmental benefit would be \$2,280 per vehicle.

With these caveats in mind, our main results show that the subsidy for electric vehicles is not justified by environmental benefits. But, as noted above, there are other arguments in favor of electric vehicle subsidies. Perhaps most important is the possibility of the long-term benefits due to a combination of innovation spillovers, learning by doing, and dynamic changes to the electricity grid. Any such long-term benefits may be at least partially offset by the short-term costs associated with current electric vehicle use. Our analysis provides an estimate of these short-term costs. Moreover, by shedding light on issues related to differentiated regulation and pollution export, we provide a policy framework for subsequent analysis of long-term issues.

In Section I we develop a simple general equilibrium model that includes discrete choice over vehicle type as well as environmental externalities from driving. In Section II we describe the methods by which we determine emissions and damages from electric and gasoline vehicles. Section III presents the results. In Section IV we discuss the caveats to our analysis. Section V concludes.

I. Theoretical Model

Consider a theoretical discrete choice transportation model in which consumers in the market for a new vehicle choose between a gasoline vehicle and an electric

[11] The emissions inventory used by our integrated assessment model (AP2) is from 2011. These are the latest years for which all data are available.

vehicle.[12] Consumers obtain utility from a composite consumption good x (with price normalized to one) and from miles driven over the life of the selected vehicle, either gasoline miles g or electric miles e. We allow for several policy variables. The government may provide a subsidy s for the purchase of an electric vehicle, place a tax t_g on gasoline miles, a tax t_e on electric miles, or some combination of these policies.[13] We hold fuel and vehicle prices fixed.[14]

The indirect utility of purchasing a gasoline vehicle is

$$V_g = \max_{x,g} x + f(g) \text{ such that } x + (p_g + t_g)g = I - p_\Psi,$$

where p_Ψ is the price of the gasoline vehicle, p_g is the price of a gasoline mile, I is income, and f is a concave function. Likewise, the indirect utility of purchasing an electric vehicle is

$$V_e = \max_{x,e} x + h(e) \text{ such that } x + (p_e + t_e)e = I - (p_\Omega - s),$$

where p_Ω is the price of the electric vehicle, p_e is the price of an electric mile, and h is a concave function. Any difference in attributes between the gasoline and electric vehicle are captured by differences in the functions f and g. Because the objective function in these optimization problems is quasi-linear, there are no income effects.[15]

Following the discrete choice literature, we assume that the choice of vehicle is influenced by i.i.d. random variables ϵ_g and ϵ_e drawn from a common extreme value distribution with zero expected value and standard deviation that is proportional to a parameter μ.[16] Accordingly, we define

$$\mathcal{U}_g = V_g + \epsilon_g,$$

and

$$\mathcal{U}_e = V_e + \epsilon_e.$$

A consumer selects the gasoline vehicle if $\mathcal{U}_g > \mathcal{U}_e$. This occurs with probability

$$\pi \equiv \text{Probability}(\mathcal{U}_g > \mathcal{U}_e) = \frac{\exp(V_g/\mu)}{\exp(V_g/\mu) + \exp(V_e/\mu)}.$$

The expected utility of a new vehicle purchase is given by

$$E\big[\max[\mathcal{U}_e, \mathcal{U}_g]\big] = \mu \ln\big(\exp(V_e/\mu) + \exp(V_g/\mu)\big).$$

[12] Examples of general discrete choice models are Anderson, Palma, and Thisse (1992) and Small and Rosen (1981). Applications to transportation models are De Borger (2001) and De Borger and Mayeres (2007). In online Appendix A, we extend the model to include several vehicles of each type.

[13] Alternatively, we might consider a tax on fuel consumption. These taxes are equivalent in our model, but may not be equivalent in a model with multiple vehicles of each type. See Fullerton and West (2002).

[14] This is consistent with a model in which vehicles and miles are produced by price-taking firms using constant returns to scale technology.

[15] The marginal utility of income is equal to one, the number of miles driven does not depend on income, and the choice of vehicle does not depend on income.

[16] The variance is $\mu^2(3.14159)^2/6$.

Consumers create negative environmental externalities by driving, but ignore the damages from these externalities when making choices about the type of vehicle and number of miles. In our empirical analysis, gasoline vehicles emit several pollutants from their tailpipes and electric vehicles cause emissions of several pollutants from the smokestacks of electric power plants that charge them. Because the damages from these pollutants may be global or local, we introduce multiple locations into the model.

A. *Uniform versus Differentiated Regulation*

Let m denote the number of locations and let α_i denote the proportion of the total population of new vehicle buyers that resides in location i. An important feature of our model is that driving in one location may lead to local damages in that location, as well as local damages in other locations. Accordingly, we define *full damages* due to driving in location i as the sum across all locations of local damages plus the global damages. Assuming that both global and local damage functions are linear allows us to characterize full damages with a single variable for each type of vehicle.[17] Let δ_{gi} denote the marginal full damages (in dollars per mile) from driving a gasoline vehicle in location i, and δ_{ei} denote the marginal full damages (in dollars per mile) from driving an electric vehicle in location i.

We determine welfare maximizing purchase subsidies under both *uniform regulation* (the same policy applies to all locations) and *differentiated regulation* (policy may vary from location to location). Because the first-best policy in our model is differentiated Pigouvian taxes on both types of miles, we refer to the welfare maximizing subsidies as second-best.[18]

First we study differentiated regulation. Here there are m local governments that select location-specific purchase subsidies. Let R_i denote the expected government revenue generated by the purchase of a new vehicle in location i.[19] For the moment, we assume local government i cares about full damages due to driving in location i. It selects the purchase subsidy s_i to maximize the welfare $\mathcal{W}_i$ associated with the purchase of a new vehicle within location i, defined as the sum of expected utility and expected revenue less expected pollution damage:[20]

$$\mathcal{W}_i = \mu\big(\ln(\exp(V_{ei}/\mu) + \exp(V_{gi}/\mu))\big) + R_i - (\delta_{gi}\pi_i g_i + \delta_{ei}(1-\pi_i)e_i).$$

Optimizing the welfare function gives the following proposition (all proofs are in online Appendix A).

[17] Constant marginal damages is consistent with the EPA's social cost of carbon calculations as well as prior research on local air pollution (Muller and Mendelsohn 2009; Fowlie and Muller 2013).

[18] Results for uniform taxes on miles are in online Appendix B.

[19] Alternatively we could have a single revenue equation and assume that a central government makes the location-specific policy choices. But, given our subsequent distinction between full and native damages, it is natural to consider distinct local governments.

[20] Because there are no income effects, the consumer component of welfare is equivalent to the standard notion of compensating variation (Small and Rosen 1981).

PROPOSITION 1: *The second-best differentiated subsidy on the purchase of the electric vehicle in location i is given by* s_i^* *where*

$$s_i^* = \left(\delta_{gi} g_i - \delta_{ei} e_i\right).$$

The term $\delta_{gi} g_i - \delta_{ei} e_i$ is simply the difference between the full damages over the driving lifetime of a gasoline vehicle and the full damages over the driving lifetime of an electric vehicle.[21] Even if the electric vehicle emits less pollution per mile than the gasoline vehicle, the sign of the subsidy is ambiguous, because the number of miles driven may be different. If the miles driven are indeed the same, and the electric vehicle emits less pollution per mile than the gasoline vehicle, then the subsidy is positive. We refer to the difference $\delta_{gi} - \delta_{ei}$ as the *environmental benefits* of an electric vehicle. This concept assumes that the number of miles driven by the two types of vehicles is the same (an assumption we will maintain in most of the empirical section below).

Next we study uniform regulation. Here a central government selects a uniform subsidy that applies to all m locations. The government's objective is to maximize $\sum \alpha_i W_i$, which is the weighted average of welfare across locations. The next proposition delineates the second-best uniform subsidy. It also describes an approximation formula for the welfare gain in moving from uniform regulation to differentiated regulation.

PROPOSITION 2: *Assume that prices, income, and the functions h and g are the same across locations. The second-best uniform subsidy on the purchase of an electric vehicle is given by* $\tilde{s}$, *where*

$$\tilde{s} = \left(\left(\sum \alpha_i \delta_{gi}\right) g - \left(\sum \alpha_i \delta_{ei}\right) e\right).$$

Furthermore, let $\mathcal{W}(S^*)$ *be the weighted average of welfare from using the second-best differentiated subsidies* s_i^* *in each location and let* $\mathcal{W}(\tilde{S})$ *be the weighted average of welfare from using the second-best uniform subsidy* $\tilde{s}$ *in each location. To a second-order approximation, we have*

$$\mathcal{W}(S^*) - \mathcal{W}(\tilde{S}) \approx \frac{1}{2}\pi(1-\pi)\left(\frac{1}{\mu}\sum \alpha_i (s_i^* - \tilde{s})^2 - \frac{1}{\mu^2}(1-2\pi)\sum \alpha_i (s_i^* - \tilde{s})^3\right),$$

where π *is evaluated at the uniform subsidy.*

These results are most easily interpreted in the special case in which the population of new vehicle buyers is the same in each location $\left(\alpha_i = \frac{1}{n}\right)$ and the electric vehicle and gasoline vehicle are driven the same number of miles ($g = e$).[22] Here the second-best uniform subsidy $\tilde{s}$ is equal to the average environmental benefits

[21] The formula for s_i^* has a simple structure because there are two vehicles in the choice set. If the choice set is larger, then s_i^* will depend on the various cross-price elasticities (see online Appendix A).

[22] To test the robustness of the results in Propositions 1 and 2, we also analyze a model in which consumers make a continuous choice between gasoline and electric miles. See online Appendix C.

multiplied by the number of miles driven. And the approximate welfare gain from differentiation is a function of the second and third moments of the distribution of the environmental benefits. To understand these results, consider marginal welfare in region i:

$$\frac{\partial \mathcal{W}_i}{\partial s_i} = \frac{\pi(1-\pi)}{\mu}\left(-s_i + g\left(\delta_{gi} - \delta_{ei}\right)\right).$$

When set equal to zero in a first-order condition, the policy variable s_i can be solved for as a linear function of the environmental benefits. But the marginal welfare function itself is a nonlinear function of s_i (through the variable π). If it had been linear in s_i, then the approximate welfare gain from differentiation would not have been a function of the third moment.[23]

Proposition 2 provides a point of comparison to previous work on differentiated regulation. But the practical application of the approximation is limited because it depends on the value of μ. Recall that this parameter is proportional to the standard deviation of the random variables in the utility function. If we determine a value for μ, either by an econometric procedure (Dubin and McFadden 1984) or by a calibration procedure (De Borger and Mayeres 2007), then we will generally be able to determine the exact numerical value of the welfare gain, which eliminates the need for an approximation.

B. *Full versus Native Damages*

So far we have assumed that local government i is concerned with the full damages caused by driving in location i. But this may not necessarily hold. For example, when an electric vehicle is driven in Pennsylvania, regulators in Pennsylvania may be more concerned about environmental damages which occur in Pennsylvania than they are about downwind damages that occur in New York. To account for this possibility, it is useful to break up full damages into *native damages* (i.e., those damages which occur in location i) and *exported damages* (i.e., those which occur in other locations.)

If a local government only cares about native damages, then its objective is to maximize

$$\hat{\mathcal{W}}_i \ = \ \mu\big(\ln(\exp(V_{ei}/\mu) \ + \ \exp(V_{gi}/\mu))\big) \ + \ R_i - (\hat{\delta}_{gi}\pi_i g_i \ + \ \hat{\delta}_{ei}(1-\pi_i)e_i),$$

where $\hat{\delta}_{gi}$ and $\hat{\delta}_{ei}$ are the marginal native damages in location i due to driving a vehicle in location i. It follows from Proposition 1 that the second-best purchase subsidy based on native damages, denoted by $\hat{s}_i^*$, is given by

$$\hat{s}_i^* \ = \ \left(\hat{\delta}_{gi}g_i - \hat{\delta}_{ei}e_i\right).$$

[23] For more details and a comparison with Mendelsohn (1986), see online Appendix D. See also Jacobsen et al. (2016).

We would expect considerable heterogeneity across locations in the relationship between native and full damages due to the various chemical and physical processes that govern the flow of local pollution. In general, however, we would expect electric vehicles to export more pollution than gasoline vehicles, due to the distributed nature of electricity generation as well as the fact that smokestacks release emissions much higher in the atmosphere than tailpipes. The greater the extent to which the electric emissions are exported to other locations, the greater the extent to which a given location may want to subsidize the purchase of an electric vehicle.

II. Calculating Air Pollution Damages

The theoretical model illustrates that the environmental benefits of an electric vehicle arises from reduced damages relative to the gasoline vehicle it replaces. We calculate this benefit by determining emissions per mile for electric and gasoline vehicles, and then mapping emissions into damages, accounting for the fact that both emissions and damages may differ by location. In these calculations, we use the county as the basic unit of location. We first give an overview of our general procedure, and then describe the details of our two component empirical models.

We consider damages from five pollutants: CO_2, SO_2, NO_x, $PM_{2.5}$, and VOCs. These pollutants account for the majority of global and local air pollution damages and have been a major focus of public policy.[24] Our set of electric vehicles includes each of the 11 pure electric vehicles in the EPA fuel efficiency database for the 2014 model year.[25] Our set of gasoline vehicles is meant to capture the closest substitute in terms of non-price attributes to each electric vehicle. Wherever possible, we use the gasoline-powered version of the identical vehicle, e.g., the gasoline-powered Ford Focus for the electric Ford Focus.[26]

To determine the emissions per mile for each gasoline vehicle, we integrate data from several sources.[27] For CO_2 and SO_2, emissions are directly proportional to gasoline usage, so we use conversion factors in the Greenhouse Gases, Regulated Emissions, and Energy Use in Transportation (GREET) model scaled.[28] We differentiate urban and nonurban counties by using the EPA's city and highway mileage.[29] For NO_x emissions, we use the Tier 2 emission standards for the vehicle "bin." For $PM_{2.5}$ and VOCs, we combine the Tier 2 standards with GREET estimates of $PM_{2.5}$ emissions from tires and brakes and VOC emissions from evaporation. The implication of this procedure is that emissions per mile for each gasoline vehicle only differ across urban and nonurban counties.[30]

[24] We do not analyze emissions of CO and toxics like mercury. Most CO emissions are from vehicles, and most toxics are from power plants, so the direction of bias from these omissions is unclear.

[25] The federal purchase subsidy depends on the size of the battery. All 11 pure electric vehicles are eligible for the maximum subsidy of $7,500.

[26] Survey data on new vehicle purchases provided by MaritzCX was used to verify that these choices were reasonable. See online Appendix E for details.

[27] We do not make state-level adjustments to car emissions, although fuel blend regulations in California have been shown to improve air quality (Auffhammer and Kellogg 2011).

[28] In the 2012 GREET model, developed by Argonne National Laboratory, the SO_2 emissions rate is 0.00616 g/mile at 23.4 mpg. This is slightly higher than the Tier 2 allowed 30 ppm which would be 0.00485 g/mile at 23.4 mpg.

[29] Urban counties are defined as counties which are part of a Metropolitan Statistical Area (MSA).

[30] The emissions per mile for our gasoline vehicles are reported in Table A in online Appendix E.

For electric vehicles, determining emissions per mile is more complicated. We begin with the EPA estimate of mpg equivalent (i.e., the estimated kWh per mile).[31] We adjust this figure to account for the temperature profile of each county, because electric vehicles use more electricity per mile in cold and hot weather.[32] Next we use an econometric model (described below) to estimate the marginal emissions factors (e.g., tons per kWh) for each of our pollutants at each of 1,486 power plants due to an increase in regional electricity load. We combine these estimates with an assumed daily charging profile to determine the emissions per mile at each power plant due to the charging of an electric vehicle in a given county.[33] The implication of this procedure is that emissions per mile for each electric vehicle may differ across any two counties.

Next we map emissions into damages. For CO_2, we use the EPA social cost of carbon of \$41 per ton.[34] For local pollutants, we use the AP2 model. This model calculates damages per unit of a given local pollutant in each county (as described below). By multiplying emissions per mile and damages per unit emitted, and then aggregating across pollutants (and, for electric vehicles, across power plants) we obtain the full damages per mile for each gasoline vehicle and each electric vehicle in each county. As in the theoretical section, these full damages account for global effects, local effects in the given county, and local effects in other counties.

To analyze any policy which affects multiple counties, we need a sense of the relative importance of driving in the counties. We weight all summary statistics using vehicle miles traveled (VMT) in each county, as estimated by the EPA for their Motor Vehicle Emission Simulator (MOVES).[35]

A. *Econometric Model: Estimation of Marginal Emission Factors from Electricity Use*

To determine the emissions that result from electricity use to charge an electric vehicle, we must determine which power plants respond (and how they respond) to increases in electricity usage at different locations. The electricity grid in the contiguous United States consists of three main "interconnections": Eastern, Western, and Texas. Since there are substantial electricity flows within each interconnection but quite limited flows between interconnections, we model each interconnection separately. Within each interconnection, transmission constraints prevent the free flow of electricity throughout the interconnection. Accordingly, we follow the North

[31] We use the combined city/highway EPA figure and do not differentiate electric vehicles by urban and rural since regenerative braking leads to smaller differences in city and highway efficiencies.

[32] This is due to both the decreased performance of the battery and the increased demand for climate control (Yuksel and Michalek 2015). Temperature has a smaller effect on the performance of gasoline vehicles, so we do not adjust gasoline mpg for temperature. We model the electric vehicle range loss as a Gaussian distribution with no range loss at 68ºF but a 33 percent range loss at 19.4ºF. See online Appendix G. We explore how sensitive our findings are to this assumption, as well as others, in Section IIID.

[33] We analyze eight charging profiles: our baseline profile using estimates from Electric Power Research Institute (EPRI) (see Figure B in online Appendix F), a flat profile, and six profiles with nonoverlapping four-hour charging blocks.

[34] See "The Social Cost of Carbon," EPA, https://www.epa.gov/climatechange/social-cost-carbon. We use the year 2015, 3 percent discount rate estimate and convert it to 2014 dollars. Moreover, all monetary values in all model components are also converted to 2014 dollars.

[35] If vehicle life and miles driven per year per vehicle are the same across counties, then these weights are equivalent to the weights α_i (the number of new vehicle buyers) used in the theoretical model.

American Electric Reliability Corporation (NERC) and divide the three interconnections into nine distinct regions.[36] We use these nine NERC regions to define the spatial scale for measuring emissions per kWh. In particular, our estimation strategy assumes that an electric vehicle charged at any county within a given NERC region has the same marginal emission factors as an electric vehicle charged at any other county within the same region.[37]

Our data consists of hourly emissions of CO_2, SO_2, NO_x, and $PM_{2.5}$ at 1,486 power plants as well as hourly electricity consumption (i.e., electricity load) for each of our nine NERC regions, for the years 2010–2012.[38] We use these data to estimate the effect of electricity load on emissions, employing methods similar to Graff Zivin, Kotchen, and Mansur (2014) and Holland and Mansur (2008). Like them, we allow for an integrated market where electricity consumed within an interconnection may be provided by any power plant within that interconnection. In contrast, however, we estimate the effect of changes in electricity load *separately* for each power plant in the interconnection.

The dependent variable in our analysis, y_{it}, is power plant i's hourly emissions (CO_2, SO_2, NO_x, or $PM_{2.5}$) at time t. For each power plant, we regress the dependent variable on the contemporaneous electricity load in each of the regions within the power plant's interconnection. To account for different charging profiles, the coefficients on load vary by hour of the day. The regression includes fixed effects for each hour of the day interacted with the month of the sample. We regress

$$y_{it} = \sum_{h=1}^{24} \sum_{j=1}^{J(i)} \beta_{ijh} HOUR_h LOAD_{jt} + \sum_{h=1}^{24} \sum_{m=1}^{36} \alpha_{ihm} HOUR_h MONTH_m + \varepsilon_{it},$$

where $J(i)$ equals the number of regions in the interconnection in which power plant i is located, $HOUR_h$ is an indicator variable for hour of the day h, $MONTH_m$ indicates month of the sample m, and $LOAD_{jt}$ is the electricity consumed in region j at time t. The coefficients of interest are the marginal emission factors β_{ijh}, which represent the change in emissions at plant i from an increase in electricity usage in region j in hour of the day h.

B. *The AP2 Model: Determining Damages from Local Air Pollution*

The AP2 model is an integrated assessment air pollution model.[39] AP2 connects reported emissions (USEPA 2014) to estimates of ambient concentrations using an air quality model. In particular, the air quality model maps emissions of ammonia, NO_x, SO_2, $PM_{2.5}$, and VOCs from each reported source of air pollution in the contiguous United States into ambient concentrations of SO_2, O_3, and $PM_{2.5}$ at all receptor locations (i.e., the 3,110 counties in the contiguous United

[36] See online Appendix H for our procedure for assigning counties to NERC regions.

[37] There are some data on electricity load at NERC subregions. Due to a high degree of multi-collinearity, our estimation strategy would likely not work at this level of disaggregation.

[38] CO_2, SO_2, and NO_x data are directly from the EPA continuous emission monitoring system (CEMS). We construct hourly $PM_{2.5}$ from hourly generation and annual $PM_{2.5}$ emissions rates. Power plant emissions of VOCs are negligible. More details about this data are in online Appendix I.

[39] See Muller (2011). More details of our implementation of AP2 are given in online Appendix I.

States). The remaining components of AP2 then link these ambient concentrations to exposures, physical effects, and monetary damages. Welfare endpoints covered by the model include: human health, crop and timber yields, degradation of buildings and material, and reduced visibility and recreation (Muller and Mendelsohn 2007). Human exposures are calculated using county-level population data for 2011 which are reported by the US Census. Crop and timber yields are reported by the US Department of Agriculture. Damages associated with built structures, visibility, and recreation contribute a very small share of total damage (Muller, Mendelsohn, and Nordhaus 2011).

Exposures are translated into physical effects (e.g., premature deaths, cases of illness, lost crop yields) using concentration-response functions reported in the related literature. In terms of the share of total damages, the most important concentration-response functions are those governing adult mortality. We use results from Pope et al. (2002) to specify the effect of $PM_{2.5}$ exposure on adult mortality rates and we use results from Bell et al. (2004) to specify the effect of O_3 exposure on all-age mortality rates.[40] Mortality risks, which comprise the vast majority of damage from local air pollution, are then expressed in terms of monetary terms using an $8.1 million value of a statistical life (VSL). Crop and timber yield effects from pollution exposure are valued using 2011 market prices.

Because of the focus of this paper on small changes to the vehicle fleet, calculation of incremental damages per-unit mass emitted is necessary. The algorithm used to compute damages per ton herein has been used in prior research (Muller and Mendelsohn 2009; Muller, Mendelsohn, and Nordhaus 2011). Briefly, this entails the following steps. With all sources in the United States emitting at their reported level in 2011, exposures, physical effects, and monetary damages are computed. Then, for an emission from a particular power plant, AP2 adds one ton of SO_2, for example, to reported emissions for 2011. Exposures, physical effects, and monetary damage are re-computed. The incremental damages per-unit mass is tabulated as the difference in monetary damage between the baseline case and the add-one-ton case.

Importantly, in computing per-unit emitted damages, AP2 aggregates the difference in damages across all county receptors affected by the additional ton. As discussed above, local governments may be more concerned about native damages rather than full damages. We use the AP2 model in a novel way to determine both types of damages. To determine full damages, we follow the usual procedure and aggregate damages at *all* receptors. To determine native damages, we only consider damages that occur at receptors within the state or county of interest. For example, driving a gasoline vehicle in Fulton County, Georgia, creates a plume of pollution concentrated in a few counties. Gasoline native damages for Fulton County correspond to the portion of the plume that harms Fulton County. Driving an electric car in Fulton County leads to an increase in emissions from various power plants, as specified by the econometric model. The resulting plumes of pollution cover a number of counties. Electric native damages for Fulton County correspond to the portions of the plumes that harm Fulton County.

[40] In our sensitivity analysis, we study a more recent concentration response function (Roman et al. 2008).

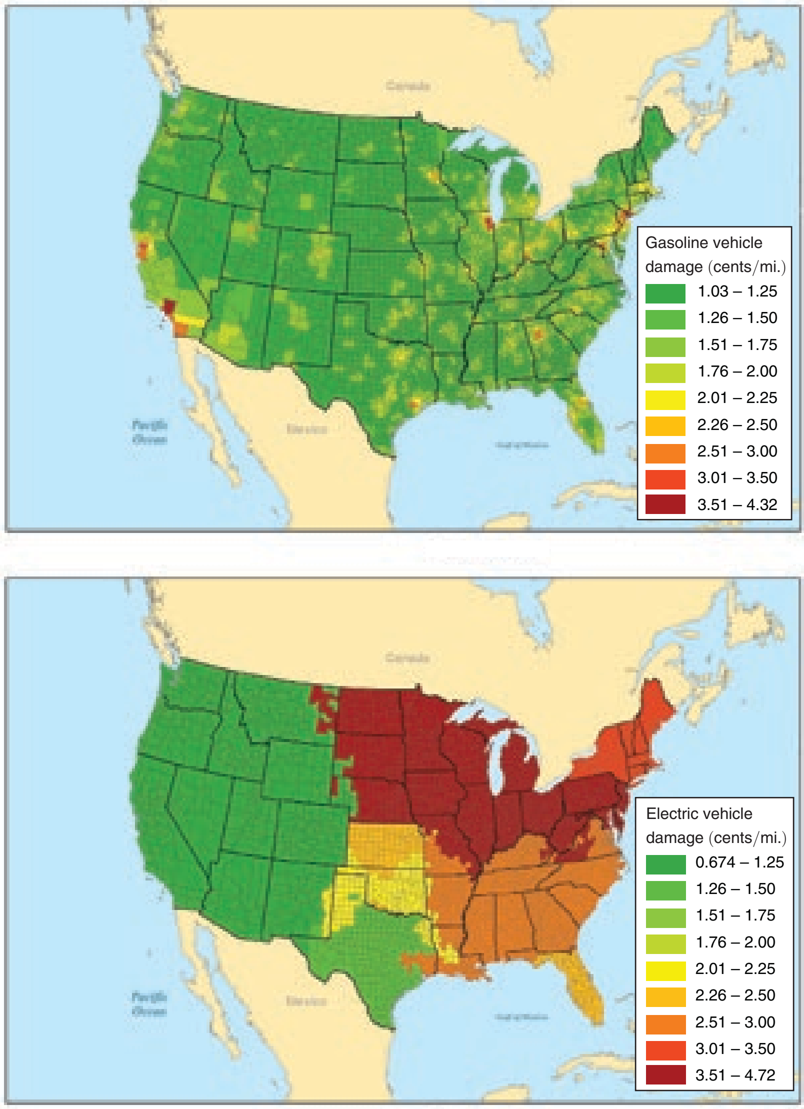

Figure 1. Marginal Damages for Gas and Electric Vehicles by County

III. Results

A. *Environmental Benefits of Electric Vehicles*

The environmental benefits of electric vehicles depend on the difference between damages from gasoline and electric vehicles. We begin with damages from electric vehicles. The bottom panel of Figure 1 illustrates our baseline estimates of the damages (in cents per mile) for the 2014 electric Ford Focus by county.[41] The variation

[41] See online Appendix Q for full page color versions of all figures.

Table 1—Mean Damages in Cents per Mile by Electricity Demand Region for a 2014 Ford Focus Electric Vehicle for Different Charging Profiles

Region	EPRI	Flat	Hr 1–4	Hr 5–8	Hr 9–12	Hr 13–16	Hr 17–20	Hr 21–24	VMT (%)
California	0.69	0.75	0.65	0.78	0.78	0.84	0.82	0.64	11
WECC w/o CA	1.03	0.92	1.18	0.98	0.84	0.76	0.73	0.99	11
ERCOT	1.28	1.21	1.50	1.41	1.10	1.07	1.05	1.16	7
SPP	2.24	2.74	2.07	4.91	2.30	2.89	2.39	1.89	4
FRCC	2.48	2.14	3.21	2.36	2.25	1.39	1.53	2.11	6
SERC	2.75	2.67	2.75	2.26	2.72	2.96	2.63	2.71	22
NPCC	3.11	2.75	4.19	3.75	1.61	2.12	2.49	2.35	9
RFC	3.64	3.55	3.42	3.38	3.83	3.06	3.43	4.15	17
MISO & MRO	4.29	3.52	5.63	3.91	3.03	2.57	2.32	3.69	14
Total	2.59	2.41	2.90	2.56	2.28	2.15	2.12	2.46	100

Notes: The regions are ordered by the damage per mile under the EPRI charging profile. The EPRI charging profile is illustrated in Figure B in online Appendix F; the flat charging profile assumes charging is equally likely across hours; and other profiles assume charging occurs only in the indicated hours. Damages (in cents per mile) are weighted across counties by passenger vehicle VMT. California is the California ISO; WECC w/o CA is the Western United States excluding California; ERCOT is Texas; SPP is Kansas and Oklahoma; FRCC is Florida; SERC is the Southeast; NPCC is the Northeast; RFC is the Mid-Atlantic and Midwest; and MISO & MRO is the upper Midwest. See Figure C in online Appendix H for a map of the regions.

is largely driven by the NERC regions, although damages do vary within a region due to our county-specific temperature correction.

Table 1 summarizes the data in Figure 1 and shows sensitivity with respect to charging profiles.[42] In the baseline EPRI profile, mean damages are 2.6 cents per mile (the equivalent of 8.1 cents per kWh) but range from \$0.01 or less per mile in California and the West (WECC) to over \$0.04 per mile in the Midwest (MISO and MRO). These regional differences in emissions reflect the pollution intensity of the fuels used in each region's generating capacity as well as its electricity imports from other regions. There is some variation in damages across the charging profiles. Our baseline results are based on the EPRI charging profile, in which most electric vehicle charging occurs at night. However, damages could be reduced in the Midwest (MISO and MRO) by over 1.5 cents per mile by charging between 1 PM and 4 PM, for example. But generally, variation across charging profiles is much smaller than the variation across NERC regions.

The left columns of panel A in Table 2 summarize the distribution of damages across counties for the electric Ford Focus as well as all other 2014 model year electric vehicles. For the electric Ford Focus, the mean is 2.59 cents per mile with a range from under \$0.01 (in the West) to almost \$0.05 (in the Midwest). The difference across vehicles is due solely to differences in their efficiency (in kWh per mile). For example, the BYD e6 (the dirtiest electric vehicle) uses approximately twice as many kWh per mile as the Chevy Spark (the cleanest electric vehicle). Correspondingly, the mean, minimum, and maximum damages of the BYD e6 are approximately double those of the Chevy Spark.

We now turn to the damages from gasoline vehicles. The top panel of Figure 1 illustrates the damages (in cents per mile) for the gasoline Ford Focus by county.

[42] All results are in 2014\$ and all summary statistics are weighted by VMT.

Table 2—Summary Statistics of Damages and Environmental Benefits in Cents per Mile for 2014 Electric Vehicles and Substitute 2014 Gasoline Vehicles across Counties

	Electric vehicle			Gasoline vehicle			Environmental benefits		
Vehicle	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
Panel A. Damages and environmental benefits									
Chevy Spark	2.28	0.59	4.17	1.69	0.95	4.30	−0.60	−3.15	3.08
Honda Fit	2.30	0.60	4.20	1.93	1.13	4.83	−0.37	−3.00	3.60
Fiat 500e	2.34	0.61	4.27	1.74	0.93	4.62	−0.60	−3.26	3.33
Nissan Leaf	2.38	0.62	4.35	1.23	0.74	3.53	−1.16	−3.52	2.22
Mitsubishi i-Miev	2.42	0.63	4.41	1.69	0.95	4.30	−0.73	−3.40	3.05
Smart fortwo	2.54	0.66	4.63	1.67	0.98	4.50	−0.87	−3.57	3.13
Ford Focus	2.59	0.67	4.72	1.86	1.03	4.32	−0.73	−3.63	3.16
Tesla S (60 kWh)	2.82	0.73	5.15	2.44	1.28	5.48	−0.38	−3.78	4.28
Tesla S (85 kWh)	3.06	0.80	5.59	2.67	1.48	5.74	−0.39	−4.02	4.55
Toyota Rav4	3.58	0.93	6.52	2.09	1.20	5.02	−1.49	−5.23	3.50
BYD e6	4.35	1.13	7.94	2.09	1.20	5.02	−2.27	−6.64	3.30
	Environmental benefits			Global environmental benefits			Local environmental benefits		
Vehicle	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
Panel B. Decomposition of environmental benefits into global and local environmental benefits									
Chevy Spark	−0.60	−3.15	3.08	0.35	−0.14	0.72	−0.95	−3.01	2.37
Honda Fit	−0.37	−3.00	3.60	0.52	0.02	0.89	−0.89	−3.02	2.71
Fiat 500e	−0.60	−3.26	3.33	0.32	−0.19	0.71	−0.92	−3.08	2.63
Nissan Leaf	−1.16	−3.52	2.22	−0.09	−0.40	0.28	−1.07	−3.16	1.99
Mitsubishi i-Miev	−0.73	−3.40	3.05	0.30	−0.21	0.69	−1.04	−3.20	2.36
Smart fortwo	−0.87	−3.57	3.13	0.18	−0.24	0.57	−1.06	−3.34	2.57
Ford Focus	−0.73	−3.63	3.16	0.44	−0.21	0.89	−1.17	−3.43	2.28
Tesla S (60 kWh)	−0.38	−3.78	4.28	0.83	-0.07	1.36	−1.21	−3.72	2.93
Tesla S (85 kWh)	−0.39	−4.02	4.55	0.96	0.01	1.54	−1.36	−4.04	3.02
Toyota Rav4	−1.49	−5.23	3.50	0.23	−0.51	0.81	−1.72	−4.73	2.71
BYD e6	−2.27	−6.64	3.30	−0.04	−0.88	0.65	−2.23	−5.78	2.66

Notes: Damages are from power plant emissions or tailpipe emissions of NO_x, VOCs, $PM_{2.5}$, SO_2, and CO_2. Electric vehicles assume the EPRI charging profile. Substitute vehicles are defined as the identical make where possible. The substitute vehicle for the Nissan Leaf is the Toyota Prius; for the Mitsubishi i-Miev is the Chevy Spark; for the Tesla Model S is the BMW 740 or 750; and for the BYD e6 is the Toyota Rav4. Damages are in cents per mile and are weighted across counties by VMT.

The counties with large damages correspond to major population centers because air pollution damages are mostly comprised of premature mortality risks. These damages are summarized in the middle columns of panel A in Table 2. For the gasoline Ford Focus, mean damages are 1.86 cents per mile (the equivalent of \$0.51 per gallon) but range from about \$0.01 per mile to over \$0.04 per mile.[43]

Notice that there is substantial overlap in the distributions of damages from gasoline and electric vehicles. If these damages were highly correlated, then the environmental benefits of an electric vehicle would be small in most counties. In fact, the damages are not highly correlated (the correlation is 0.07). As a result, the environmental benefits vary substantially, as shown in the right columns of panel A in Table 2. For example, gasoline vehicle damages are large in Los Angeles (due to the large population and properties of the airshed) but electric vehicle damages are small (due to the clean Western power grid). In this situation, the environmental benefits are almost equal to gasoline damages (i.e., \$0.03 to \$0.04 per mile) and hence electric vehicles have substantial environmental benefits. The opposite occurs in the upper Midwest where gasoline vehicle damages are small (due to low

[43] Mean damages per gallon of gasoline range from \$0.48 to \$0.62 across the vehicles. For the Ford Focus, damages across counties range from \$0.37 to \$1.12 per gallon.

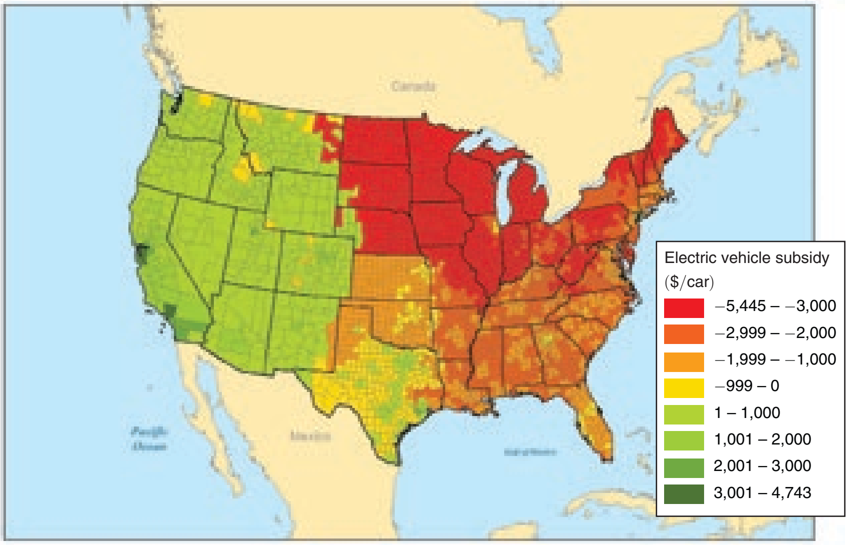

Figure 2. Second-Best Electric Vehicle Subsidy by County

population densities) but electric vehicle damages are large (due to the prevalence of coal-fired generation in the region and the temperature adjustment to electric vehicle range). Here the environmental benefits of an electric vehicle are *negative*, and are almost equal to the electric vehicle damages. Overall, the environmental benefits are negative on average for each of the electric vehicles in Table 2, panel A.[44] The electric Ford Focus is the median electric vehicle in terms of environmental benefits, and we focus on it throughout the results section.

Panel B in Table 2 decomposes the environmental benefits into global benefits and local benefits. Just about every electric vehicle, in just about every place, creates global environmental benefits relative to gasoline vehicles. In contrast, the local environmental benefits from electric vehicles can be positive or negative depending on the place. But on average, for all electric vehicles, the negative local environmental benefits outweighs the positive global environmental benefits. Focusing solely on global environmental benefits provides a misleading impression of the environmental consequences of electric vehicles.[45]

Using Proposition 2, we can convert the environmental benefits into the second-best purchase subsidy by assuming that both the electric vehicle and the gasoline vehicle are driven 150,000 miles.[46] Figure 2 shows the second-best subsidies by county. Except for a few counties around New York City and Atlanta, the subsidy is negative throughout the eastern part of the country (i.e., it is a *tax* on the purchase

[44] This is due in large part to the fact that only 30 percent of the VMT occurs in the three regions with the lowest marginal damages from electricity (see the last column of Table 1).

[45] Several prominent online sites that compare gasoline and electric vehicles (EPA, Union of Concerned Scientists) only consider global environmental benefits.

[46] We assume both vehicles have ten year lifetimes, regardless of the number of miles driven, and that both are driven 15,000 miles a year in the absence of any taxes on miles. In practice, vehicle life depends on both years and miles driven. Moreover, it is not clear whether electric vehicles will be driven more (due to lower costs per mile) or less (due to the inconvenience of charging) than gasoline vehicles.

Table 3—Environmental Benefits in Cents per Mile by Metropolitan Statistical Areas for a 2014 Ford Focus (*Electric versus Gasoline*)

Metropolitan statistical area	Environmental benefits per mile	VMT (percent)	Damage per mile (gasoline)	Damage per mile (electric)	Purchase subsidy
Highest benefit MSAs					
Los Angeles, CA	3.16	2.69	3.85	0.69	\$4,743
Oakland, CA	2.21	0.75	2.89	0.68	\$3,315
San Jose, CA	2.11	0.54	2.80	0.69	\$3,166
San Francisco, CA	1.91	0.45	2.59	0.68	\$2,867
Santa Ana, CA	1.87	0.93	2.54	0.67	\$2,800
Other high VMT MSAs					
San Diego, CA	1.85	0.97	2.53	0.68	\$2,770
Riverside, CA	1.17	1.35	1.88	0.71	\$1,756
Phoenix, AZ	0.74	1.16	1.77	1.03	\$1,112
Houston, TX	0.67	1.74	2.01	1.35	\$1,003
Dallas, TX	0.62	1.52	1.91	1.29	\$926
New York, NY	−0.02	1.97	3.16	3.18	−\$32
Atlanta, GA	−0.36	1.92	2.38	2.73	−\$535
Chicago, IL	−0.74	1.75	2.98	3.72	−\$1,116
Washington, DC-VA	−0.89	1.40	2.19	3.08	−\$1,335
Minneapolis, MN	−2.39	1.06	2.08	4.46	−\$3,578
US and nonurban					
US average	−0.73	100	1.86	2.59	−\$1,095
Nonurban	−1.67	20	1.20	2.87	−\$2,500
Lowest benefit MSAs					
St. Cloud, MN	−2.87	0.08	1.62	4.49	−\$4,310
Bismarck, ND	−2.97	0.04	1.52	4.49	−\$4,456
Fargo, ND-MN	−3.07	0.07	1.54	4.61	−\$4,605
Duluth, MN-WI	−3.09	0.10	1.47	4.56	−\$4,635
Grand Forks, ND-MN	−3.14	0.03	1.52	4.66	−\$4,711

Notes: The environmental benefits are the difference in damages between the gasoline-powered Ford Focus and the electric Ford Focus. Environmental benefits are weighted by VMT by county within each MSA. Nonurban includes all counties that are not part of an MSA. The purchase subsidy assumes vehicle is driven 150,000 miles.

of electric vehicles). The subsidy is large and negative in the upper Midwest. On the other hand, it is positive in most places in the West, and quite large in many counties in California. Overall, the second-best subsidy ranges from about positive \$5,000 to negative \$5,000.

In Table 3, we aggregate to the level of Metropolitan Statistical Area (MSA). The MSAs with the highest environmental benefits are all in California because electricity generation in the West does not produce much air pollution. In these MSAs, the environmental benefits are about \$0.02 to \$0.03 per mile (a second-best subsidy of up to \$5,000). The MSAs with the lowest environmental benefits are all in the upper Midwest, again because of the prevalence of coal-fired power stations. Here the environmental benefits are −\$0.03 per mile (a second-best purchase tax of about \$4,000). Other large MSAs can have either positive or negative environmental benefits. New York and Chicago have some of the largest damages from gasoline vehicles, but environmental benefits from electric vehicles are small or negative due to the large damages from electric vehicles. Electric vehicles have substantial environmental benefits in the major Texas MSAs, due to relatively low electric

Table 4—Environmental Benefits in Cents per Mile by State for a 2014 Ford Focus (*Electric versus Gasoline*)

State	Environmental benefits per mile	VMT (percent)	Damage per mile (gasoline)	Damage per mile (electric)	Purchase subsidy
Highest benefit states					
California	1.86	11	2.55	0.69	$2,785
Utah	0.73	1	1.77	1.04	$1,089
Colorado	0.60	2	1.63	1.03	$902
Arizona	0.59	2	1.62	1.02	$889
Washington	0.58	2	1.59	1.02	$865
Other high VMT states					
Texas	0.34	8	1.75	1.41	$505
Florida	−0.70	7	1.80	2.49	−$1,049
Georgia	−0.78	4	1.96	2.74	−$1,166
New York	−0.91	5	2.19	3.10	−$1,371
North Carolina	−1.07	4	1.67	2.74	−$1,611
Virginia	−1.20	3	1.72	2.93	−$1,807
Illinois	−1.56	3	2.31	3.87	−$2,345
Ohio	−1.76	4	1.89	3.65	−$2,640
Pennsylvania	−1.78	3	1.86	3.64	−$2,675
Michigan	−2.48	3	1.76	4.24	−$3,720
Lowest benefit states					
South Dakota	−2.66	0	1.27	3.93	−$3,992
Minnesota	−2.76	2	1.72	4.48	−$4,145
Wisconsin	−2.79	2	1.59	4.37	−$4,180
Iowa	−2.93	1	1.37	4.30	−$4,394
North Dakota	−3.31	0	1.27	4.58	−$4,964
US average	−0.73	100	1.86	2.59	−$1,095

Notes: The environmental benefits are the difference in damages between the gasoline-powered Ford Focus and the electric Ford Focus. Environmental benefits are weighted by gasoline-vehicle VMT within each state. The purchase subsidy assumes the vehicle is driven 150,000 miles.

vehicle damages in Texas. However, for nonurban regions as well as for MSAs in the Southeast, Northeast, and Midwest, the benefits from electric vehicles are negative.

Table 4 contains a similar analysis at the state level. Compared to MSAs, the environmental benefits of electric vehicles are smaller at the state level because of negative benefits in nonurban areas. The largest environmental benefits are in California (a second-best subsidy of $3,000) and other Western states. The lowest benefits are in the upper Midwest (a second-best tax of almost $5,000 in North Dakota). There are only 11 states in which the environmental benefits are positive, and Texas is the only high VMT state outside the Western interconnection in which the environmental benefits are positive. The top panel of Figure 3 shows the second-best purchase subsidy by state. When driven in the average state, a 2014 electric Ford Focus causes $1,095 more environmental damages over its driving lifetime than the equivalent gasoline Ford Focus.[47]

[47] Although our main focus is on variation from this average, and the main focus in Michalek et al. (2011) is on life-cycle costs, we can compare our results to theirs. They find, on average, a battery electric vehicle causes $181 more environmental damages over its driving lifetime than a gasoline vehicle. See online Appendix O for details.

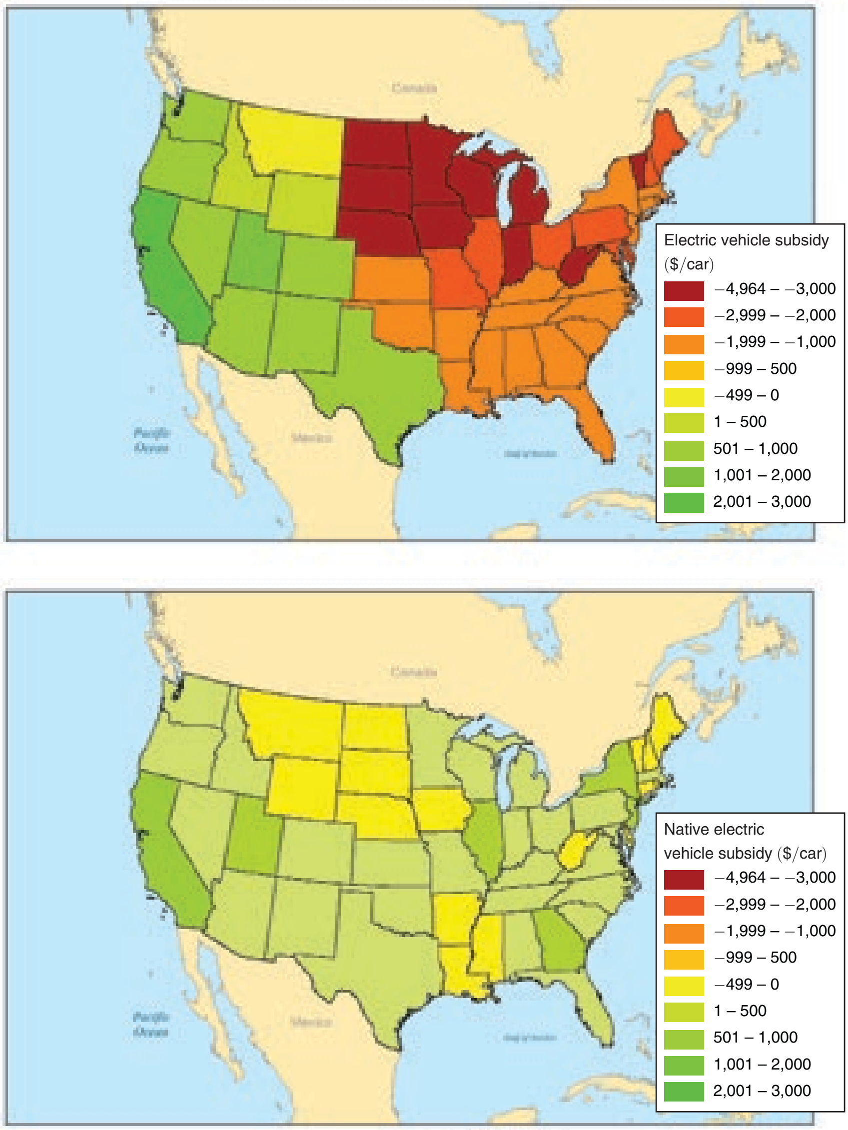

Figure 3. Second-Best Electric Vehicle Subsidy by State (*Full and Native Damages*)

B. *Exporting Pollution: Full and Native Damages*

Although both gasoline and electric vehicles export pollution, electric vehicles export pollution to a remarkable degree (the grid itself is distributed and emissions from power plants are released from tall smokestacks intended to disperse pollutants over a wide area).[48] To illustrate this discrepancy, we first analyze transport of a specific pollutant from a specific county. The top panel in Figure 4 illustrates the change in $PM_{2.5}$ associated with driving gasoline-powered Ford Focus vehicles

[48] Similarly, any electricity consuming good will export pollution.

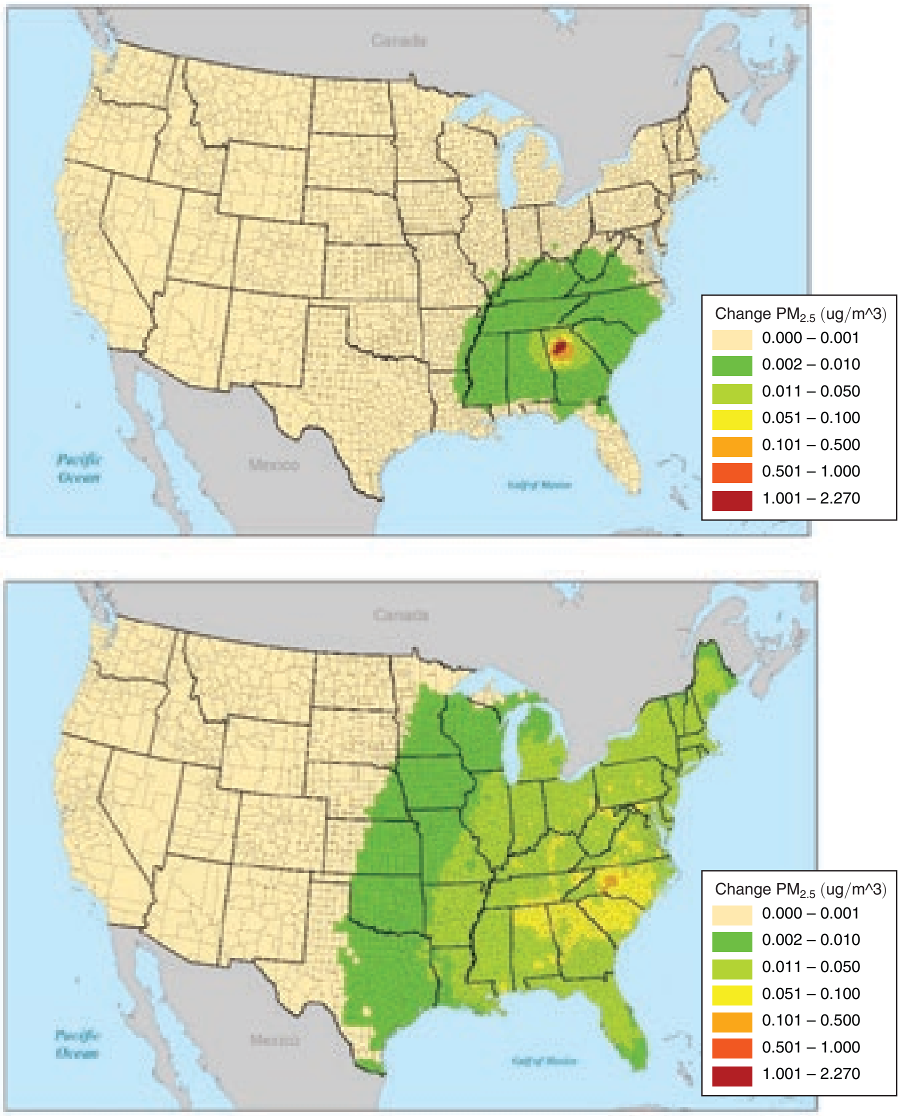

FIGURE 4. CHANGE IN $PM_{2.5}$ FROM GASOLINE VERSUS ELECTRIC VEHICLE IN FULTON COUNTY, GEORGIA

Notes: The top panel illustrates the change in $PM_{2.5}$ associated with a fleet of 10,000 gasoline-powdered Ford Focus vehicles, each driven 15,000 miles in a year in Fulton County. The bottom panel illustrates the change in $PM_{2.5}$ associated with the same number of miles driven by electric powered Ford Focus vehicles charged in Fulton County, thereby increasing the consumption of electricity in the United States (SERC).

in Fulton County, Georgia. Most of the increase in $PM_{2.5}$ is centered within a few nearby counties. The bottom panel in Figure 4 shows the change in $PM_{2.5}$ associated with equivalent driving by electric powered Ford Focus vehicles charged in the same county. The spatial footprint of $PM_{2.5}$ in this case encompasses the entire eastern United States.[49]

[49] In the online supplementary material, we present similar maps for the 20 largest MSAs.

Table 5—Native Damages in Cents per Mile by State and County and Export Percentages

Vehicle	Damages	Mean	Med.	SD	Min	Max
Electric	All	2.59	2.74	1.18	0.67	4.72
	Non-GHG	1.70	1.86	1.02	0.16	3.50
	State	0.15	0.16	0.07	0.04	0.33
	Export percent	91	91			91
	County	0.02	0.02	0.01	0.00	0.06
	Export percent	99	99			98
Gasoline	All	1.86	1.76	0.59	1.03	4.32
	Non-GHG	0.53	0.36	0.52	0.01	2.92
	State	0.43	0.26	0.51	0.00	2.76
	Export percent	19	28			5
	County	0.23	0.10	0.37	0.00	2.03
	Export percent	57	72			30
Environmental benefits	All	−0.73	−1.01	1.39	−3.63	3.16
	Non-GHG	−1.17	−1.48	1.19	−3.43	2.28
	State	0.28	0.12	0.51	−0.32	2.46
	County	0.21	0.08	0.37	−0.06	2.00

Notes: Damages in cents per mile. All reports damages from all pollutants at all receptors. Non-GHG reports damages from local pollutants (i.e., excluding CO_2) at all receptors. State reports damages from local pollutants from receptors within the same state as the source. County reports damages from local pollutants from receptors within the same county as the source. State Export percent reports the share of non-GHG damages which occur at receptors outside the state. County Export percent reports the share of non-GHG damages which occur at receptors outside the county. Electric damages assume the EPRI charging profile. Damages are weighted by VMT.

Our definition of native damages allows a more comprehensive analysis of pollution export. Table 5 shows native damages at both the state and county levels for both electric and gasoline vehicles. For electric vehicles, full damages from local pollutants are 1.7 cents per mile on average. Native state damages are only 0.15 cents per mile, and native county damages are only 0.02 cents per mile. Thus on average 91 percent of electric vehicle damages from local pollutants are exported from the state and 99 percent are exported from the county. Local damages from gasoline vehicles are exported to a much smaller extent. On average only 19 percent of these damages are exported from a state and only 57 percent are exported from a county.

Using native damages rather than full damages changes the environmental benefits calculation quite dramatically, especially at the lower tail of the distribution. In this lower tail, gasoline full damages are small and electric full damages are large. Because most electric vehicle damages are exported, both native gasoline damages and native electric damages are small. This implies that the lower tail of environmental benefits moves from approximately -3.6 cents per mile to approximately -0.06 cents per mile for county-level native damages. In contrast, at the upper tail of the distribution, electric vehicle damages were already low, so accounting for native damages has a smaller impact on the environmental benefits. On average, the environmental benefits calculated using native damages is positive at both the state and county level. Correspondingly, as illustrated in the bottom panel of Figure 3, the state level second-best purchase subsidy, using native damages, is positive in 32 out of 48 states.

Do state policymakers place greater emphasis on full or native damages when considering electric vehicle subsidies? A number of states have implemented subsidies for the adoption of electric vehicles, above and beyond the federal subsidy, such as California ($2,500), Colorado ($6,000), Georgia ($5,000), Illinois ($4,000), and Maryland ($3,000). In addition, some states offer a variety of other incentives, including carpool lane access, electricity discounts, and parking benefits.[50] As shown in online Appendix J, both actual subsidies and the number of other incentives are more highly correlated with our calculated native damage subsidy than with our calculated full damage subsidy. This evidence suggests that native damages may help explain state policymakers' support for electric vehicle subsidies.

C. *State and County Differentiated Policies*

Our analysis shows that the environmental benefits of electric vehicles vary substantially across locations. This raises the question of whether differentiated policies can lead to large enough welfare gains to offset any additional implementation costs. To illustrate these welfare gains, we calibrate the discrete choice model developed in Section II.[51] In addition to electric vehicle purchase subsidies, we also consider fuel-specific taxes on miles driven (i.e., VMT taxes), because such taxes at the county level correspond to first-best policy in our model. Some care must be taken with the interpretation of these results, because they rely on a number of assumptions and specific functional forms. Nevertheless, they illustrate the complexity of comparing the gains from differentiation across different policies.

Panel A in Table 6 shows the deadweight losses for differentiated VMT tax policies. County-specific taxes on electric miles and gasoline miles set at the Pigovian levels $t_{ei} = \delta_{ei}$ and $t_{gi} = \delta_{gi}$ have zero deadweight loss. To calculate deadweight losses of other policies, we need to specify the share of new vehicle purchases that would be electric under a default policy in which there is no subsidy at all (or business as usual). If the share would be 2 percent, we refer to this as the 2 percent BAU EV share case. We consider 1 percent, 2 percent, and 5 percent shares. Even at a 5 percent share, the electricity used to charge the vehicles is a small percentage of the overall variation in electricity load (see online Appendix M). Given a 2 percent BAU EV share, state-specific taxes have a deadweight loss of $92 million per year, and uniform federal taxes have a deadweight loss of $191 million per year.[52] This implies a gain from differentiation of $100 million (moving from federal to state) and of $191 million (moving from federal to county). The middle and right columns of panel A in Table 6 show differentiated policies in which there is only a single tax on one of the fuels. The second-best single tax is smaller than the Pigovian tax, because consumers can avoid taxation by substituting into the untaxed vehicle (see online Appendix L). For single tax policies, the gains from differentiation are on the order of $25–$200 million. However, the deadweight losses are large particularly for taxes on electric miles only ($1.7 billion). The last three rows of panel A in Table 6

[50] The Department of Energy maintains a database of alternative fuels policies by state (http://www.afdc.energy.gov/laws/matrix?sort_by=tech). A few states impose a special registration fee for electric vehicles. Our data accounts for policies in place on July 28, 2014.

[51] See online Appendix K for more details.

[52] For context, annual vehicle sales are approximately 15 million in the United States.

Table 6—Deadweight Losses of Differentiated VMT Taxes and Differentiated Purchase Subsidies

Panel A. Deadweight losses of differentiated VMT taxes	Gas and electric tax BAU EV share percent			Gas tax only BAU EV share percent			Electric tax only BAU EV share percent		
	1	2	5	1	2	5	1	2	5
County policies	0	0	0	201	391	905	1,709	1,717	1,740
State policies	89	92	102	289	482	1,005	1,712	1,721	1,752
Federal policies	162	191	277	343	542	1,095	1,736	1,770	1,874
County (native)	989	1,073	1,325						
State (native)	1,067	1,153	1,412						
Federal (native)	778	809	903						

Panel B. Deadweight losses of differentiated electric vehicle purchase subsidies	BAU EV share percent		
	1	2	5
County policies	1,754	1,806	1,960
State policies	1,758	1,815	1,983
Federal policies (−$1,095 subsidy)	1,783	1,864	2,107
County policies (native damages)	1,788	1,874	2,134
State policies (native damages)	1,792	1,881	2,152
Federal policies (native damages)	1,785	1,868	2,188
Current federal policy ($7,500 subsidy)	2,581	3,459	6,079
BAU federal policy (zero subsidy)	1,791	1,880	2,148

Notes: Deadweight loss in millions of dollars per year is based on 15 million annual vehicle sales normalized to the emissions profile of the Ford Focus. The BAU EV share is the proportion of electric vehicles sold if there were no subsidy. This share is determined by the assumed value for μ (10,664, 10,508, 10,037) which is proportional to the standard deviation of the unobserved relative preference shock. In panel A, federal taxes in the joint tax case are 1.9 cents per mile on gasoline miles and 2.6 cents per mile on electric miles.

show differentiated taxes based on native damages. The gains from differentiation are small or even negative. These policies lead to large deadweight losses ($0.7–$1.5 billion), because taxes based on native damages are much too low.

Panel B in Table 6 shows the deadweight losses for differentiated electric vehicle purchase subsidies. Gains from differentiation are relatively small: on the order of $10–$60 million at a 2 percent BAU EV share. These gains are much smaller than the gains from differentiation of VMT taxes. The distribution of environmental benefits is right skewed. Because the probability of adopting the gas vehicle is close to one, it follows from Proposition 2 that this skewness leads to an increase in the gains from differentiation. Deadweight losses from electric vehicle subsidies are large: around $1.8 billion per year. Electric vehicle subsidies based on native damages have similarly large deadweight losses and small gains from differentiation.

Finally, panel B in Table 6 shows the deadweight loss from the current federal policy of a $7,500 subsidy on the purchase on an electric vehicle and the deadweight loss from the default no-subsidy policy. The deadweight loss from the current federal subsidy is $3.4 billion per year at a 2 percent BAU EV share. This exceeds the deadweight loss from the no-subsidy policy by $1.6 billion per year. The BAU EV shares shown in the table represent plausible shares in the near future and are appropriate for evaluating policy looking forward. To evaluate the recent past, we calculate deadweight losses of the two policies for a BAU EV share of 0.375 percent which is consistent with the actual 2014 electric vehicle market share of approximately

0.75 percent.[53] The deadweight loss from the current federal subsidy is \$2.0 billion and the deadweight loss from the no-subsidy policy is \$1.7 billion. Regardless of BAU EV share, the current federal subsidy has larger deadweight loss than the no subsidy policy. And the welfare difference between the two polices increases substantially as the BAU EV share increases.

D. *Sensitivity Analysis*

Our analysis takes data from a number of different sources, uses estimated coefficients from regressions in the electricity model and the AP2 model, and makes assumptions about variables such as charging behavior and the effects of temperature on electric vehicle range. Although there is uncertainty associated with each of these factors, we do not attempt to assign standard errors to our results. Instead we perform a sensitivity analysis to see the effects of various deviations from our baseline model.[54]

The first parameter that we explore in Table 7 is the social cost of carbon (SCC). Our baseline value is \$41. A higher value for the SCC leads to higher damage estimates for both electric and gasoline vehicles, but the environmental benefits are not highly sensitive to the assumed SCC.

Several of our assumptions affect only one type of vehicle. On the electric side, our baseline calculation makes a temperature adjustment to account for the reduced performance of electric vehicles in weather extremes and uses the EPRI charging profile. Table 7 shows that our results are not sensitive to these choices. On the gasoline side, our baseline calculation differentiates the mpg of gasoline vehicles by city and highway driving and assumes emissions throughout the lifetime of the vehicle are the same as when new. Using an average mpg instead leads to slightly lower gasoline vehicle damages. Doubling emissions rates for local pollutants primarily affects the upper tail of the gasoline vehicle damages and hence the upper tail of the environmental benefits.

Another set of assumptions relate to parameters in the AP2 model. In particular, in the baseline case, AP2 uses a VSL of approximately \$8.1 million. A lower VSL of about \$2.7 million leads to lower damages for both electric and gasoline vehicles and hence a narrower distribution for the environmental benefits. Another important parameter in AP2 is the dose-response function that links $PM_{2.5}$ exposure to adult mortality. We find that a higher dose response parameter leads to higher damages for both vehicles which widens the distribution of environmental benefits.

The next calculation examines changes to the electricity grid and the gasoline vehicle fleet. Our baseline uses observed power plant emissions in 2010–2012 to estimate the damages from electric vehicles. New air pollution and climate regulations on power plants will likely lead to lower emissions in the future. In addition, there is an ongoing transition from coal plants to gas plants. For a rough estimate of these effects, we model a power grid in which all of the coal-fired power plants are replaced with new gas-fired power plants. This procedure implies that the replacement plants would be in the same locations and would be dispatched identically to

[53] Li et al. (forthcoming) estimate that 50 percent of electric vehicle sales are due to the subsidy.
[54] Additional sensitivity for the welfare analysis is in online Appendix K.

Table 7—Sensitivity Analysis of Damages and Environmental Benefits in Cents per Mile for 2014 Electric and Gasoline Ford Focus

	Electric vehicle			Gasoline vehicle			Environmental benefits		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
Baseline	2.59	0.67	4.72	1.86	1.03	4.32	−0.73	−3.63	3.16
Carbon cost									
SCC = \$51	2.80	0.80	5.02	2.18	1.28	4.67	−0.62	−3.68	3.38
SCC = \$31	2.37	0.55	4.42	1.53	0.78	3.98	−0.84	−3.58	2.95
No temperature adjustment	2.43	0.67	3.90	1.86	1.03	4.32	−0.57	−2.84	3.18
Flat charging profile	2.41	0.74	3.88	1.86	1.03	4.32	−0.55	−2.79	3.10
Average mpg	2.59	0.67	4.72	1.74	1.23	4.11	−0.85	−3.42	2.89
Double gasoline emission rates	2.59	0.67	4.72	2.39	1.05	7.24	−0.20	−3.58	5.60
\$2.7 Million VSL	1.61	0.71	2.64	1.54	1.02	2.55	−0.06	−1.59	1.63
PM dose response	3.74	1.25	6.89	2.16	1.04	5.96	−1.58	−5.76	3.91
Future grid & vehicle	0.67	0.37	1.39	1.23	0.74	3.53	0.56	−0.57	2.73
High estimates ∗	2.65	0.69	4.46	2.13	1.15	3.90	−0.52	−2.86	2.36
Low estimates ∗	2.53	0.68	4.15	1.59	0.78	2.90	−0.94	−3.12	1.87

Notes: Baseline corresponds to Ford Focus row from panel A in Table 2. Carbon cost uses a social cost of carbon of \$51 or \$31. No temperature adjustment assumes electric vehicle range is independent of temperature. Flat charging profile assumes electric vehicle charging occurs equally in all hours. Average mpg uses the average mpg for the gasoline vehicle regardless of where it is driven. Double gasoline emissions rates doubles the gasoline vehicle emissions rates for local pollutants. \$2.7 Million VSL assumes the VSL is \$2.7 million instead of the baseline \$8.1 million. PM dose response assumes a higher $PM_{2.5}$ adult-mortality dose-response from Roman et al. (2008). Future grid & vehicle assumes all coal-fired power plants replaced by identically dispatched natural gas plants and a Toyota Prius gasoline vehicle. High estimates assumes ninety-fifth percentile damages for all local pollutants for all counties. Low estimates assumes fifth percentile damages for all local pollutants for all counties. ∗ indicates the min and max counties are replaced by the fifth and ninety-fifth percentile counties.

the old coal-fired plants.[55] Turning to the gasoline vehicle fleet, our baseline uses the gasoline Ford Focus as the comparison vehicle to the electric Ford Focus. New regulations on gasoline vehicles will likely lead to lower emissions in the future. For a rough estimate of these effects, we use the Toyota Prius as a proxy for the vehicle of the future. The effect of these changes on the environmental benefits of electric vehicles is given by the "Future grid and vehicle" row in Table 7. Damages from both vehicles are lower, and damages from electric vehicles are much lower. However the mean environmental benefits of 0.56 cents per mile implies an electric vehicle subsidy of \$840, which is still substantially less than current subsidies.

Finally, we consider statistical uncertainty associated with the marginal damages produced by AP2 for both gas and electric vehicles. The procedure is described in online Appendix I. The results using the fifth and ninety-fifth percentiles for the damages are reported in Table 7.

IV. Caveats and Other Considerations

There are several important caveats to our calculation of the environmental benefits of an electric vehicle due to decreased air pollution emissions. First, we

[55] Modeling different plant locations and a new load curve is beyond the scope of the present analysis. Here we scale the plant-specific coefficients for coal plants by a ratio. The numerator is the average emissions rate for combined cycle gas turbine plants that started operating after 2007, namely their total emissions in 2010 over their total net generation that year. The denominator is a similar emissions rate for each coal plant in our sample that is not a co-generation plant.

have only considered air pollution emissions associated with driving the vehicles. There are other "upstream" environmental externalities associated with electric and gasoline vehicles.[56] It is unlikely, however, that these upstream externalities have the same degree of heterogeneity found in the air pollution emissions from driving. So the effect of including them would likely be a shift in the distribution of second-best subsidies but not a significant change in the variance of this distribution. Previous research has shown that electric vehicles have approximately $1,500 greater upstream externalities than gasoline vehicles (Michalek et al. 2011).[57]

Second, our analysis is based on a simple snapshot of the electricity grid in the years 2010–2012. We might expect the grid to become cleaner over time by integrating new lower-emission fuels and technologies. Of course, gasoline vehicles may become cleaner over time as well. The overall effect on the environmental benefits of electric vehicles will depend on the relative rates of changes of these two factors. Table 7 has an analysis of a future grid, but it is important to stress that our estimates are based on the dispatch and emissions of the electricity grid in 2010–2012.

Third, we focus on the marginal emissions from an increase in the demand for electric power due to electric vehicles charging. This is appropriate when the electricity demand for electric vehicles is a small fraction of overall electricity use. In online Appendix M, we discuss large scale adoption of electric vehicles.

Fourth, we analyze the environmental benefits of electric vehicles in isolation from other environmental regulations. In practice, these regulations may impact the market for vehicles and/or the electricity market, and hence have an effect on the environmental benefits of electric vehicles.[58]

Some regulations will have a negative effect on the environmental benefits of electric vehicles. Consider the Corporate Average Fuel Economy (CAFE) standards. CAFE stipulates that the sales-weighted harmonic mean of mpg for a given manufacturer's fleet of vehicles must meet a certain requirement. Electric vehicles are assigned a mpg value for this calculation. These values are much larger than any existing gasoline vehicle. Assuming that the CAFE requirement is initially binding, selling an electric vehicle enables manufacturers to meet a lower standard for the rest of their fleet. Let the CAFE-induced environmental cost of an electric vehicle be defined as the increase in environmental damage from the rest of the fleet when an electric vehicle is sold. In online Appendix N, we determine the CAFE-induced environmental cost and show that the second-best subsidy on the purchase of an electric vehicle is decreased by the amount of the CAFE-induced environmental cost. Applying our baseline values for the Ford Focus, the CAFE-induced environmental cost is $1,555 per vehicle.[59] This value is significant in comparison with even the largest second-best subsidy for an electric vehicle found in our study ($2,785, in California).

[56] These include emissions from making vehicles and batteries, extracting oil, refining gasoline, transporting gasoline to retail stations, mining coal and natural gas, and transporting these resources to electric plants.

[57] See online Appendix O. See also Tamayao et al. (2015) and the references therein.

[58] In addition, there may be preexisting distortions in both the electricity market (e.g., regulatory pricing policy) and the gasoline market (e.g., OPEC).

[59] A more thorough analysis would use a complete model of both supply and demand for the entire new vehicle market and relax our assumption of constant prices. See also Jenn, Azevedo, and Michalek (2016).

Other regulations, such as cap-and-trade programs and renewable portfolio standards (RPS), will have a positive effect on the environmental benefits. EPA programs cap emissions of NO_x and SO_2, and the Regional Greenhouse Gas Initiative caps emissions of CO_2 in the Northeast. In our model of the electricity market, we determine the marginal increase in emissions due to an increase in electricity consumption. We do not model the constraint that power plant emissions are capped. During the period of our analysis, permit prices were exceedingly low in many markets, especially those for SO_2. In all permit markets, the stock of banked allowances was increasing significantly despite low prices. This suggests that the cap may not have been binding in these markets.[60] Nevertheless, in online Appendix P, we perform calculations to approximate the effect of binding caps. Under the assumption that caps on NO_x, SO_x, and CO_2 are all binding, damages from an electric vehicle decrease from 2.59 cents per mile to 0.94 cents per mile (with 92 percent of this decrease due to the cap on SO_2). Equivalently, the second-best subsidy increases from −\$1,095 to \$1,380. Turning to RPS, these programs require a fixed percentage of electricity be produced by low emission technologies such as solar and wind. In a region with a RPS, an increase in the electricity load will result in a increase in low emission generation. Therefore, electric vehicle damages can be scaled by $1 - R$, where R is the RPS share, if the renewables operate at the same time and location as EV charging.

In addition to the environmental benefits studied in our paper, there are a variety of other considerations that are put forth in favor of electric vehicle subsidies. First, reducing the consumption of oil may generate geopolitical benefits, reduced military expenditures, and economic benefits from insulation to oil price shocks. Michalek et al. (2011) determined these benefits to be approximately \$1,400. Notice that this number has about the same magnitude, but the opposite sign, as the difference in upstream externalities between electric and gasoline vehicles.

Second, electric vehicle subsidies may be justified due to innovation spillovers. If innovation is a public good, then markets may provide too little innovation. Similarly, the inability of firms to appropriate the full gains from innovation (e.g., consumers may also benefit) may reduce innovation incentives. Our analysis cannot speak to the appropriateness of these justifications for electric vehicle subsidies. However, it is worth noting that electric vehicle subsidies are a "demand pull" innovation policy and hence are subject to all the limitations of demand pull policies (Jaffe, Newell, and Stavins 2005).

Third, subsidizing electric vehicles today helps boost demand, which in turn increases incentives to provide electric vehicle charging infrastructure.[61] The increase in demand may also lead to lower production costs in the future due to learning by doing. Both of these effects increase adoption in the future, which will presumably be desirable due to a cleaner electric grid. This argument may indeed have merit, but any such long-term benefits may be at least partially offset by the short-term costs associated with current electric vehicle use. Our analysis provides an estimate of these costs.

[60] A nonbinding cap may still yield positive permit prices due to transactions costs.

[61] Li et al. (forthcoming) examine the relative effectiveness of the current policy with alternative policies aimed at building out the charging network.

V. Conclusion

The comparison of environmental externalities from driving gasoline and electric vehicles depends critically on damages from local pollution. Ignoring local pollution leads to an overestimate of the benefits of electric vehicles and an underestimate of the geographic heterogeneity. Accounting for both global and local pollution, we find electric vehicles generate negative environmental benefits of 0.73 cents per mile on average relative to comparable gasoline vehicles. There is considerable variation around this average: electric vehicles used in Los Angeles, California produce benefits of 3.2 cents per mile while those used in Grand Forks, North Dakota produce benefits of −3.1 cents per mile. On average, electric vehicles driven in metropolitan areas generate benefits of about $0.01 per mile while those driven outside metropolitan areas generate benefits of −1.7 cents per mile.

These findings raise questions regarding the sign, the magnitude, and the one-size-fits-all nature of the uniform federal subsidy of $7,500 for purchasing a pure electric vehicle. Our results imply subsidies of −$1,095 on average with a range from $2,785 in California to −$4,964 in North Dakota. Thus environmental benefits from driving cannot, alone, justify the federal subsidy. As discussed above, other studies have estimated upstream environmental benefits of electric vehicles of about −$1,500 and have estimated benefits of $1,400 due to reduced oil consumption. Combining these three factors cannot justify the federal subsidy. It remains an open question as to whether or not additional considerations (such as innovation spillovers, network effects, or learning by doing) generate enough benefits to justify the federal subsidy.

At first blush, our finding of significant geographic heterogeneity in benefits suggests a need for local discretion. However, the pollution export phenomena we identify calls into question whether or not local regulation would be effective. In most states, when a consumer opts for an electric vehicle rather than a gasoline vehicle, they reduce air pollution in their state. However, in all but 11 states, this purchase makes society as a whole worse off because electric vehicles tend to export air pollution to other states more than gasoline vehicles. Given this, states may implement subsidies even though a tax might be more appropriate. Hence there may be a need for federal policy to account for exported damages.

This suggests the appropriate policy for electric vehicles should be at the federal level, but differentiated by location. We find that differentiated Pigovian taxes on miles lead to greater welfare gains than differentiated subsidies on vehicle purchases. This is not surprising, as economists have long recognized the superiority of putting a direct price on externalities relative to other indirect corrective policies. Unfortunately, this insight does not seem to have had much influence on policy, as political decision makers often implement indirect policies instead. A consequence of this predilection is that multiple indirect policies may target the same externalities, as is the case with CAFE standards and purchase subsidies on electric vehicles. In our analysis, the interaction of these policies lead additional costs of $1,555 per vehicle.

Public policy evaluation is especially difficult and important in contexts characterized by: (i) strong prior beliefs as to the merits of the policy and/or its targeted outcome; (ii) complex interactions among economic and physical systems; and

(iii) economically significant outcomes. The federal policy which encourages the purchase of electric vehicles exhibits each of these traits. Although we have focused on vehicles, there is a broader trend toward electrification of a variety of forms of transportation. Our methodology, which combines discrete-choice models, distributed electricity generation, and air pollution models, may yield a useful template for further analysis of the environmental consequences of this trend.

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